# *CprE/SE 4910 WEEKLY REPORT 03*

***09/30/2026 – 10/06/2026***

***Group number: sdmay27-14***

***Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator***

***Client: Buildertrend***

***Faculty Advisor: Dr. Islam***

***Team Members/Role:***

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

## ***Weekly Summary***

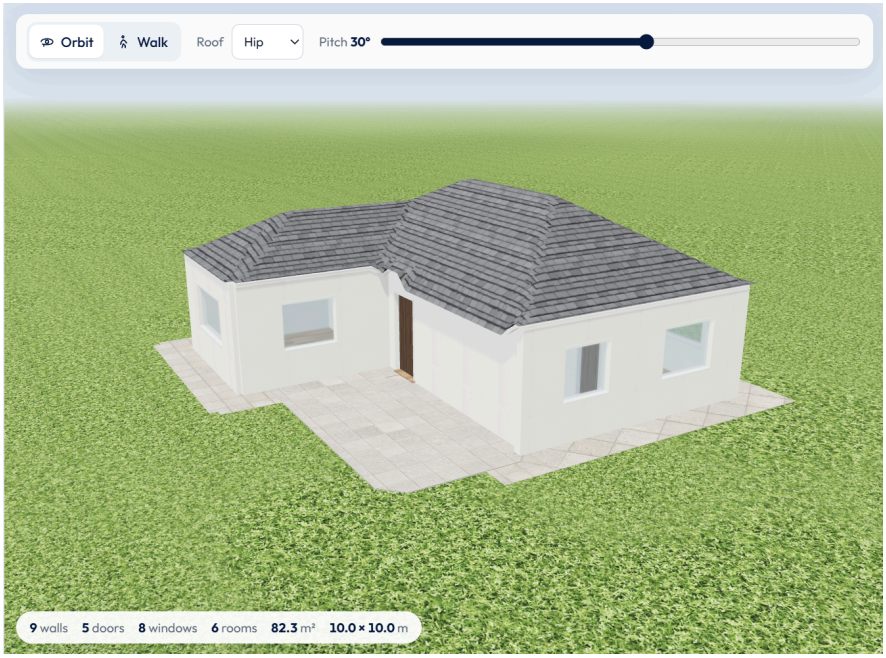
Every member of the group made small contributions to research and worked small parts of the project and prototype. There are significant limitations currently due to not having enough information from Buildertrend. Without feedback, further iterations are becoming less useful, and the more into theory the team will dive into. Significant contributions can be made once that happens.

## ***Past Week Accomplishments***

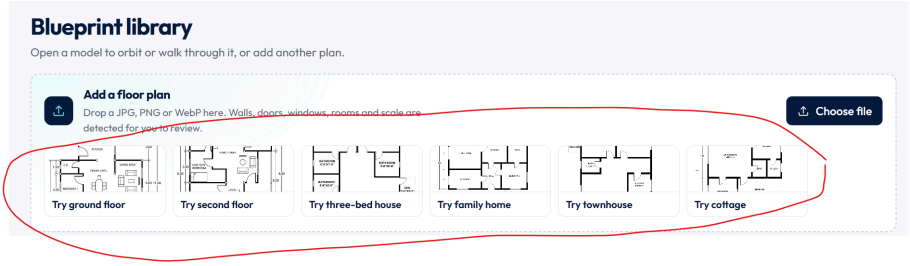
- Group Work:
    - Mostly did work related to each individuals' part of the project. Added some small contributions to the project and prototype.
- Nakshatra Gupta:
    - Continued serving as Communication Lead. Followed up with the Buildertrend contact regarding the sample blueprints and elevations that were promised and the internal developer mentor assignment.
    - Reviewed the Week 2 progress of the other members against the draft shared schema. Ryan's prototype already detects stairs, scale, and multiple roof styles (hip, gable, flat), so the schema needs fields for stairs, a scale/calibration value, and roof style options. These were logged as revisions to the draft.
    - Noted that the graphics engine is being written in C++20, so the schema must be

readable outside of Python. Began outlining a JSON Schema definition so the Python and C++ sides can validate against the same description.

- Rishiraj Parmar:
    - Researched how multi-story buildings could be represented, including floor elevation, story height, and vertical alignment between floors.
- Ryan McNab:
    - Once again made some improvements to the prototype. The computer vision was improved on more data samples, by gathering 2d blueprints from Google and Pinterest. Used simple materials for the walls, floors, roof, doors, and grass, all AI generated by a local LLM.




    - Included more example houses to the prototype, which all are generated perfectly.




    - Improved the gable roofing system to give another option of roofing, but it is still not perfect.



- Daniel Hargrave:
    - Was more limited with what could be worked on this week due to not meeting with the client or receiving any floor plans for reference. Researched a small amount.
        - Researched a bit more on NLP and some aspects with project documents. This paper shows some aspects of how NLP could be implemented with the project system.
        - NLP-based approach for automated safety requirements information retrieval from project documents
- Jase Beaubien:
    - Did more research on graphics and planning.

## ***Pending issues***

- Nakshatra Gupta:
    - The schema has only been tested against hand-written sample data. Sample plans from the client are still needed before it can be checked against realistic input. As for temporary testing, the example plans in Ryan's prototype could be used as test data.
- Rishiraj Parmar: N/A
- Ryan McNab:
    - The algorithm used to create the prototype model is still rough on new examples. Without a given format, the algorithm is plateauing since it is trying to work for all types.
- Daniel Hargrave:
    - Once again, a set of sample building plans are really, really needed. Knowing what types of floor plans that exist along with different symbols, text, and diagrams will be key in deciding how the project continues on the computer vision side. It has a really dramatic effect on both the planning and execution of this project. Until it is received, research is one of the only things that can be done on the computer vision side.

- Jase Beaubien: N/A

## ***Individual Contributions***

| **NAME** | **Individual Contributions** *(Quick list of contributions. This should be short.)* | **Hours this week** | **HOURS cumulative** |
|---|---|---|---|
| Nakshatra Gupta | Followed up with the client on sample plans and mentor assignment; reviewed team progress against the draft schema and logged revisions (stairs, scale, roof styles); began a language-neutral JSON Schema definition. | 3 | 12 |
| Rishiraj Parmar | Researched how multi-story buildings could be represented, including floor elevation, story height, and vertical alignment between floors. | 4 | 12 |
| Ryan McNab | Improved upon the prototype in many ways to help with computer vision on more examples. Also included materials for different meshes including walls, doors, roofs, floors, and grass. | 4 | 15 |
| Daniel Hargrave | Researched a small amount for a few more aspects of what the project could be like. The biggest thing that was looked into was NLP. | 3 | 13 |
| Jase Beaubien | Planning and research for graphics. | 2 | 12 |

## ***Comments and Extended Discussion***

The Professor and TA created a discord channel to discuss ongoing issues with communication to Buildertrend.

## ***Plans for the upcoming week***

- Nakshatra Gupta:
    - Work with Rishiraj to align the schema's story structure with the multi-story research (floor elevation, story height, vertical alignment).
    - Add stairs, scale, and roof style fields to the schema and finish the JSON Schema file.
    - Continue following up with the client on sample plans and the mentor assignment.
- Rishiraj Parmar:
    - Work with Nakshatra to align the schema's story structure with the multi-story research (floor elevation, story height, vertical alignment).
- Ryan McNab:
    - If feedback and samples are given, start implementing the real project based on the prototype. If not, gathering more samples and upgrading the computer vision will be the next task.
- Daniel Hargrave:
    - This largely depends on if the blueprints are given or not.
        - If affirmative, work on the model will begin.
        - Otherwise more research and setting up some environment stuff will be done.
    - Figuring out whether the project uses Yolov11? It might be good to start off using the model initially but it would cost money if Buildertrend decides to use it for commercial/company purposes. Initially, it can be used just to test everything out, then later in the class there can be a transition to another model that is either made from scratch or doesn’t cost any money. Couldn’t figure out the pricing if it would be in the final product, it's hard to say.
- Jase Beaubien:
    - Further discuss and coordinate the object format for the deserialized plans.